

Teaming Needs Scoreboard: Expert–AI Collaboration in Finance Is Delegation, Not Teamwork

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Abstract

“Human–AI teaming” has become the default vocabulary for expert–AI collaboration, and is increasingly used to justify design and procurement decisions in domains far from the crews it was built on. We found that the construct carries three unstated preconditions: jointly attributable outcomes, feedback fast enough to calibrate reliance within the working life of the collaboration, and accountability that can genuinely be shared. We test them against expert financial judgment, a domain that adopts AI advisors at scale and, unusually, audits every decision and every outcome. First, we define the calibration horizon τ , the calendar time before outcome evidence alone can establish that an AI advisor helps, and evaluate it across twelve professions using power analysis and the statistics of Sharpe-ratio inference. It ranges from about a week in flight operations to 196 years for a discretionary equity portfolio manager, while a two-hundred-manager firm running a randomised-withholding programme reaches the same evidence in about four. Identification is available to the institution and not to the individual, which inverts the relation the construct assumes, since its prescriptions are addressed to teammates. Second, we show that the dyad is the limited unit of analysis. When M experts share one advisor, privately optimal reliance cuts each expert’s squared error by 67% while making the consensus 22× worse, and we derive a correlation budget of roughly $M/4$ independently-erring models. We conclude that in this domain “teaming” describes the work less accurately than it redistributes, in language, an accountability that law makes non-delegable, and we propose delegation under agency, whose four primitives are contractible rather than psychological.

Keywords

human–AI teaming, delegation, agency theory, finance

1 Introduction

Researchers, designers and policymakers increasingly describe expert work with AI systems as “human–AI teaming”. The term organises a US National Academies consensus study [24], and it shapes how AI capabilities are specified, purchased and justified. Its appeal is that it says more than “tool”: it implies that the person and the system coordinate with each other, and it carries expectations about who decides, who answers for the result, and how much trust is warranted.

The term was developed for crews: aircrews, surgical teams and military command posts [25], and it entered human–automation research from there [19]. Those settings share several conditions that are seldom stated, because in a cockpit or an operating theatre they are never in question. The team can see what its own actions produced, it can see this quickly, and responsibility for the

result is already shared. When the term is applied to other kinds of work, these conditions are rarely re-examined, but the recommendations that rest on them are carried over unchanged. The main such recommendation is *appropriate reliance*: an expert should trust an AI system in proportion to how good that system actually is, and should learn the right proportion from experience of working with it [21].

This paper asks whether an expert can in fact follow that recommendation, in a setting where the answer can be computed rather than debated. Expert financial judgment (portfolio management, equity research, underwriting, deal advisory) is a good test case: it has used machine advisors at scale for a decade, it records every decision and every realised outcome, it has a mature statistical literature on telling skill from luck, and it assigns liability to named individuals by law. If the conditions behind the teaming vocabulary do not hold somewhere, they should be visible here, and measurable.

The paper makes three contributions:

- (1) **Preconditions, and an arithmetic test for one of them.** We make explicit three conditions the teaming construct inherits from its validating domains, then define the calibration horizon τ and evaluate it across twelve professions. The spread is nearly four orders of magnitude. We ground the effect sizes in published top-journal estimates rather than assumption, validate the closed form against outcome processes that violate its assumptions, and show that the conclusion turns on $\sqrt{T}$ rather than on parameter choices.
- (2) **Two structural results.** Identification is unit-dependent: the same effect that is unidentifiable to an expert over a career is identifiable to her employer in a few years. That dissolves an apparent paradox, namely a research literature reporting complementarity results no practitioner could have discovered, and relocates the design problem from the dyad to the institution. Separately, the dyad is the limited unit of analysis: even perfectly Bayesian individual reliance degrades collective accuracy, which yields a correlation budget on model diversity and an exact condition under which the effect reverses.
- (3) **A replacement vocabulary.** We propose delegation under agency, show which half of classical contract theory survives contact with a machine that has no preferences and no assets, and price the monitoring programme the argument implies.

2 Related Work

2.1 The teaming construct and appropriate reliance

The human-factors lineage supplies the coordination vocabulary: a team is an interdependent set of individuals with a common valued goal and assigned roles, characterised by mutual performance monitoring, backup behaviour, adaptability, shared mental models and mutual trust [25]. Klein et al. carried that vocabulary into human-automation research as a design programme [19], and the National Academies consolidated it into a research agenda [24]. The construct is contested from within: Gonzalez et al. answer the metaphor critiques by arguing that a science of human-AI teaming is achievable, and offer a complementarity framework for it [13].

The programme's prescriptive core is appropriate reliance: trust should track capability, and is calibrated through accumulated experience of the system's performance [21]. The empirical record is mixed. Meta-analytically, human-AI combinations often underperform the better of their two parts on decision tasks, with gains concentrated in content creation rather than judgment [26]. Bansal et al. supply the sharpest mechanism for why: team performance requires the human to learn the AI's error boundary, and non-parsimonious boundaries make this difficult [3]. Ben-Michael et al. give a statistical framework for the three-way comparison between human alone, human with AI, and AI alone [4].

We differ from the metaphor critiques in being conditional and falsifiable: rather than argue that the frame is universally inapplicable, we name the conditions under which it applies and give an arithmetic test for whether a domain meets them. Our claim is also the temporal complement of Bansal et al.: even a learnable error boundary cannot be learned if the evidence arrives too slowly.

2.2 AI advisors in finance, and correlated reliance

Machine advisors now measurably outperform human analysts on several tasks, and hybrids outperform both: Cao et al. report an AI analyst beating 53.7% of human target-price forecasts and a 'man + machine' hybrid beating 57.3% [5]. But reliance does not track accuracy. Auditors propose *smaller* adjustments to management's estimates when the contradictory evidence comes from their firm's AI than from a human specialist [6]: reliance tracks the source, not the signal.

Reliance on a shared model is also a population-level phenomenon. Kleinberg and Raghavan show that algorithmic monoculture can reduce welfare even when the shared algorithm is more accurate [20]. The closest antecedent to our second study is Fügener et al., who show experimentally that AI advice makes individuals more accurate while making the population more homogeneous, destroying unique human knowledge [11]. In economics the mechanism is older still: a more precise public signal can lower welfare when agents act on it in common [23]. Financial supervisors now name correlated reliance on common models and providers a stability vulnerability [10]. To this literature we add a closed-form design parameter, namely how many independently-erring models a population of given size may rely on, together with the exact condition under which shared advice helps rather than harms.

2.3 Delegation, agency, and accountability

Economics has a precise theory of delegated judgment under asymmetric information, including which signals belong in a contract [17] and when a principal actually cedes authority [2]. Empirically, human delegation to algorithms is often suboptimal: human-AI pairs beat the standalone AI only when the AI delegates to the human, because people lack metaknowledge about their own competence [12]. On accountability, Elish's moral crumple zone describes blame concentrating on the operator nearest the system [7], and human-oversight requirements are found to legitimate systems without constraining them [14]. Our contribution here is structural rather than empirical: in fiduciary settings the accountability was never transferable in the first place, so "the team decided" is a conceptual mismatch before it is an oversight problem.

3 What the Construct Presupposes

In business settings the construct arrives through two lineages at once. The human-factors definition [25] supplies the coordination vocabulary; management practice inherits Katzenbach and Smith, for whom a team is a small group holding itself mutually accountable. The second is where the accountability claim enters, and it is what a business audience hears in the word. Conflating the two is how a coordination construct acquires an accountability implication it never argued for.

Crews share three properties that the canonical dimensions presuppose.

P1. Joint, attributable outcomes. The outcome is caused mostly by the team rather than by the environment, so a record of outcomes is informative about the team. An aircrew's approach is stabilised or it is not.

P2. Resolvable feedback. Outcomes arrive fast enough, and with high enough signal-to-noise, that members can learn one another's competence during the collaboration. This is the precondition behind the whole appropriate-reliance programme [21].

P3. Shareable accountability. Responsibility genuinely distributes. "The crew erred" is a coherent finding with institutional consequences.

These were satisfied where the construct was built. The National Academies study was commissioned by the Air Force Research Laboratory for multi-domain command and control [24]: a shared mission outcome, feedback in seconds, and a command structure that already distributes responsibility. Our argument is not with the construct in that setting, but with its extension beyond it.

P2 has a distinguished ancestor. Kahneman and Klein, resolving a long disagreement about expert intuition, concluded that skill is acquirable only in environments of sufficiently high validity and with adequate opportunity to learn the regularities through rapid, unambiguous feedback [18]; they named stock picking as the canonical low-validity case. Our observation is that reliance is itself a skill, and inherits both conditions. Where an expert cannot learn to forecast, she cannot learn *from her own experience* whom to trust about forecasting. The two conditions govern skill acquired implicitly from feedback, not knowledge supplied by a designed evaluation, and that distinction becomes this paper's design recommendation rather than a loophole in its argument.

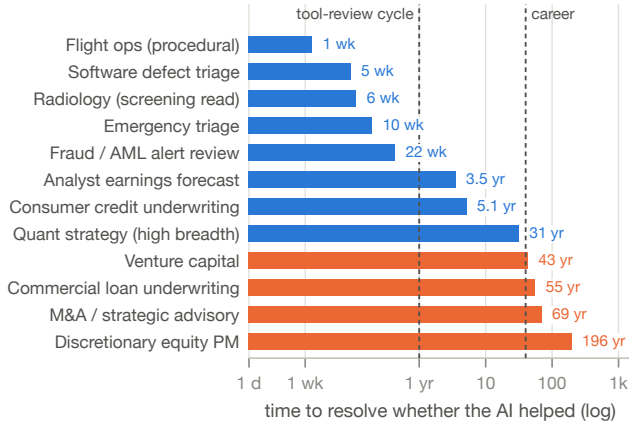


Figure 1: Calibration horizon τ : time before outcome evidence alone can establish that an AI advisor helps ($\alpha = .05$, power = .80). Blue resolves inside a 40-year career; orange does not. The four fastest domains are those teaming theory was imported from; the fifth is a finance task with a fast, resolving label. The frame follows the feedback structure, not the industry.

The preconditions are necessary but not sufficient: radiology satisfies all three tolerably well and complementarity is still elusive there, as we discuss below. In finance, P1 and P2 cease to hold together, and for one reason. Outcomes are dominated by market-wide shocks rather than by the decision, which is what makes the per-episode effect size small, so our first study treats the two as one. P1 ceases to hold a second time at population scale, which is the subject of our second study, and P3 is a legal question, which we take up when we turn to delegation.

4 Study 1: The Calibration Horizon

4.1 Model

“Calibrate your trust to the system’s actual competence” is well posed only if actual competence is estimable from the evidence the expert receives. Treat each decision as an episode. Let d be the standardised per-episode improvement in the outcome the expert is accountable for, r the episodes per year whose outcome ever resolves, and L the resolution lag. For a paired within-expert comparison at significance α and power $1 - \beta$,

$$N^* = \frac{(z_{1-\alpha/2} + z_{1-\beta})^2}{d^2}, \quad \tau = \frac{N^*}{r} + L. \quad (1)$$

We call τ the *calibration horizon*.

For portfolio decisions the natural currency is the information ratio. Let ΔIR be the annualised information ratio of the *difference* portfolio, the assisted book minus the unaided book, so that it is a Cohen’s d per year and $t = \Delta IR \sqrt{T}$ exactly [22], giving $\tau = (z_{1-\alpha/2} + z_{1-\beta})^2 / \Delta IR^2$ years. Note what this does not assume. A manager makes hundreds of position decisions a year, but because those positions share market-wide risk they yield far fewer independent observations of her skill. Breadth raises ΔIR rather than

Table 1: Domain parameters and the resulting calibration horizon. d is the standardised per-episode benefit of AI assistance on the accountable outcome; r the episodes per year whose outcome resolves; L the resolution lag. Portfolio rows use $d = \Delta IR$ and $r = 1$. Where a range was defensible we took the value most favourable to fast calibration.

Domain	d	r	L	N^*	τ (yr)
Flight ops (procedural)	0.40	2000	0.00	49	0.02
Software defect triage	0.30	1200	0.02	87	0.09
Radiology (screening read)	0.20	6000	0.08	196	0.11
Emergency triage	0.15	2000	0.02	349	0.19
Fraud / AML alert review	0.12	3000	0.25	545	0.43
Analyst earnings forecast	0.20	60	0.25	196	3.52
Consumer credit underwriting	0.05	1000	2.00	3,140	5.14
Quant strategy (high breadth)	0.50	1	0.00	31	31.40
Venture capital	0.15	10	8.00	349	42.88
Commercial loan underwriting	0.05	60	3.00	3,140	55.33
M&A / strategic advisory	0.20	3	4.00	196	69.41
Discretionary equity PM	0.20	1	0.00	196	196.22

Table 2: Realised power at N^* when the outcome process violates Eq. 1’s assumptions ($d = 0.20$, $N^* = 196$, 20,000 replications, target power 0.80). Only serial correlation matters, and it makes τ longer.

Outcome process	Power	τ factor
Gaussian (Eq. 1’s own model)	0.792	1.0×
Fat tails (t_4 , excess kurtosis 6)	0.803	1.0×
Negative skew (gamma, $\gamma_1 = -1$)	0.781	1.0×
AR(1), $\phi = 0.1$	0.772	1.2×
AR(1), $\phi = 0.2$	0.757	1.5×

the observation count [15], which is why $r = 1$ for the portfolio rows.

4.2 Why attribution and identification are one condition

P1 is often treated as a separate condition from P2, and in crews it is. In finance the two collapse into one, and it is worth seeing why exactly. Write the accountable outcome of episode i as

$$o_i = \beta f_i + \delta a_i + u_i, \quad (2)$$

where f_i is a market-wide factor, a_i indicates AI assistance with effect δ , and u_i is idiosyncratic. The standardised per-episode effect is then $d = \delta / \sqrt{\beta^2 \text{var}(f) + \text{var}(u)} = d_0 \sqrt{1 - R^2}$, where R^2 is the share of outcome variance the environment supplies and $d_0 = \delta / \text{sd}(u)$ is the effect against idiosyncratic noise alone. Because $\tau \propto 1/d^2$,

$$\tau = \frac{\tau_0}{1 - R^2}. \quad (3)$$

A limit on attribution is therefore a limit on identification, at a rate set by R^2 . For active US equity mandates, 4–8% tracking error against roughly 20% total volatility implies $R^2 \in [0.84, 0.96]$, so measuring an advisor on total outcomes rather than benchmark-adjusted ones would inflate every horizon by a further 6–25×. This

is why we work in information-ratio space throughout: Table 1 already grants finance its most favourable measurement basis, and the horizons are still centuries.

4.3 Parameters

Table 1 lists the parameters. They are order-of-magnitude estimates chosen to be generous to fast calibration: where a range was defensible we took the value most favourable to the teaming frame. The non-finance rows are illustrative and serve only as the comparison class; the finance rows carry the argument, and are examined against published estimates below.

4.4 Validation and robustness

Eq. 1 is a normal approximation, so simulating from its own generative model would establish only that we can do algebra. The informative check is to simulate outcome processes it does not assume (Table 2). Fat tails and negative skew leave realised power at the nominal 0.80, because with N^* in the hundreds or thousands the central limit theorem dominates. Serial correlation is the exception that matters: AR(1) at $\phi = 0.2$ drops realised power to 0.75 and inflates τ by roughly 1.5 $\times$. Since financial returns are autocorrelated, Eq. 1 is if anything optimistic.

4.5 Results: four orders of magnitude

Figure 1 plots τ . A line pilot resolves whether an aid helps in about a week; a screening radiologist in six. Not all of finance is slow. Fraud and AML alert review (0.4 yr), earnings forecasting (3.5) and consumer underwriting (5.1) all resolve inside a career, because their accountable outcome is adjudicated or high-frequency rather than realised through a market. The condition does not hold for delegated judgment whose outcome is a market return. A discretionary equity portfolio manager needs 196 years; an M&A adviser 69; a commercial loan officer 55; a venture capitalist 43. Even a high-breadth quantitative strategist needs 31 years, and that is the most favourable such case, since breadth is the lever the fundamental law of active management identifies [15]. Fastest to slowest spans a factor of roughly 8,000.

4.6 Are the parameters generous?

Two published anchors say yes. Cao et al.'s hybrid beats 57.3% of analyst forecasts [5]; that assisted margin, not the machine-alone rate, is what Eq. 1 needs, and it puts the analyst row at 6.3 years against the 3.5 we assume. Second, Fama and French put the cross-sectional standard deviation of true annual alpha near 1.25% [9]. Divided by the 4–8% tracking error typical of active mandates, which is the matching denominator since their alpha is benchmark-adjusted, that is an information-ratio dispersion of 0.16–0.31. Our assumed $\Delta IR = 0.20$ therefore already posits an advisor worth about one standard deviation of the whole distribution of genuine manager skill, and buys 196 years even so.

Two amplifiers make it worse. Experts evaluate several tools, so a family-wise correction inflates every horizon by 1.5–1.7 $\times$ at five to ten candidates, and finance's factor-discovery literature recommends $t = 3.0$, worth 1.9 $\times$ [16]. And the unaided counterfactual is rarely observed, so even the paired design of Eq. 1 is unavailable in practice.

The conclusion is not parameter-free. Since τ is inverse-quadratic in ΔIR , a book would be calibratable inside a 40-year career if assistance added $\Delta IR \geq 0.44$, roughly twice the total information ratio of most successful active managers. That threshold is the claim that matters, and the one most open to challenge. What *is* parameter-free is the aggregation: evidence accumulates as $\sqrt{T}$ while the effect enters linearly, so reslicing the same book into more and smaller decisions leaves τ exactly unchanged. A manager cannot buy identification by trading more often.

4.7 Identification is unit-dependent

Here is the asymmetry that matters, and it is the reason a large research literature can report complementarity results no individual practitioner could ever have discovered. N^* episodes accumulate at r per year for one expert and at $U \cdot r$ for U pooled experts, so the discretionary manager's 196 years becomes 3.9 years across a fifty-manager firm and under a year across two hundred. That is the ideal bound, and we price the practical version when we cost the monitoring programme, but the direction is what matters. Fama and French needed 3,156 funds to characterise the skill distribution [9]; no manager in that sample could characterise her own.

Pooling is not free: without randomised assignment the comparison is between self-selected adopters and non-adopters, and the advisor supplies one common error draw, so manager-level observations are not independent either. Firm-scale identification has to be designed. But that is the point. The quantity the teaming literature asks the practitioner to calibrate is estimable, just never by her and never on her timescale, which is an argument for a delegation apparatus rather than a teammate.

4.8 What experts calibrate on instead

Whatever trust financial experts form is therefore not calibrated to outcomes. It is formed from what a working week supplies: fluency, internal consistency, narrative plausibility and confident presentation, which are the surface properties generative systems optimise and which bear no established relation to accuracy. The consequence is visible in the one financial setting where it has been measured directly. Auditors propose smaller adjustments when contradictory evidence comes from their firm's AI than from a human specialist [6]: reliance tracked the source, not the signal, because the signal's quality was not observable on the timescale of the judgment.

That fast feedback is necessary but not sufficient is worth stating plainly. Screening radiology has a six-week horizon, and radiologists shown AI predictions still do not improve on average [1]; the meta-analytic picture across domains is similar [26]. Our claim is therefore one-directional: where τ exceeds a career, the reliance prescriptions cannot be executed even in principle, whatever else may also be at issue.

Some financial signals do resolve quickly, such as quarterly forecast errors and backtests, but these are proximal proxies rather than the accountable outcome, and are exactly where multiple testing bites [16]. Optimising against them substitutes a measurable sub-task for the unmeasurable judgment the mandate concerns: fast feedback in finance is available precisely where it is least trustworthy.

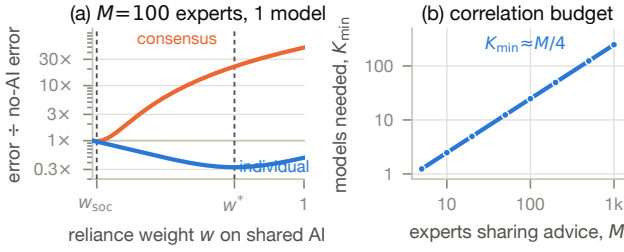


Figure 2: (a) As reliance w rises, each expert's error falls while the consensus error rises; w^* exceeds the pure-consensus optimum w_{soc} by 34 $\times$ at $M = 100$. (b) The correlation budget.

5 Study 2: Correlated Delegation

5.1 Model

Teaming is a dyadic frame; markets are not. Let M experts each hold a private signal $x_i = \theta + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0, \sigma_x^2)$ i.i.d., and let a shared advisor supply $y = \theta + \eta$, $\eta \sim \mathcal{N}(0, \sigma_y^2)$, where η is the same draw for everyone: one model, one corpus, one blind spot. Expert i reports $a_i = (1 - w)x_i + wy$. Then

$$\mathbb{E}(a_i - \theta)^2 = (1 - w)^2 \sigma_x^2 + w^2 \sigma_y^2, \quad w^* = \frac{\sigma_x^2}{\sigma_x^2 + \sigma_y^2}, \quad (4)$$

$$\mathbb{E}(\bar{a} - \theta)^2 = \frac{(1 - w)^2 \sigma_x^2}{M} + w^2 \sigma_y^2, \quad w_{\text{soc}} = \frac{\sigma_x^2}{\sigma_x^2 + M \sigma_y^2}. \quad (5)$$

Here $\bar{a}$ is the consensus that a committee vote, a supervisor's aggregate exposure, or (with capital weights in place of equal ones) a market price depends on. Accordingly M is effective breadth, the inverse Herfindahl of decision weight rather than headcount.

5.2 The private–collective wedge

Private error averages away across experts; the advisor's does not, because nobody drew an independent copy of it. Take an advisor genuinely better than any individual, $\sigma_y = 0.7\sigma_x$. Each expert is right to lean on it: at $w^* = 0.67$ her squared error falls 67%. The consensus becomes 22 $\times$ worse than if nobody had used it (Fig. 2a), and the wedge w^*/w_{soc} is 34 $\times$, growing linearly in M . We stress that w_{soc} is the pure-consensus optimum, at which each expert captures under 6% of the accuracy gain available to her. No planner holds that objective, which is why the instrument we propose below is model diversity rather than a cap on reliance. No dyad-level calibration detects any of this, because there is no dyad-level error to detect. This is the second way in which P1 does not hold: the cost is attributable to the population, and no dyad's record contains it. Closed forms and Monte Carlo agree to three decimals throughout.

5.3 When does the effect bite?

The 22 $\times$ is the pure case in which experts err independently. If private errors are correlated at ρ , a shared advisor at w^* degrades the consensus if and only if $\rho < \sigma_y^2/(\sigma_x^2 + 2\sigma_y^2)$, which is 0.25 in our calibration and, as it happens, exactly the $K_{\min}/M$ below. Simulation confirms the sign flip at that point. The claim is therefore falsifiable and cheap to test: shared advice harms the aggregate only to the extent that the model's error is more common across experts than

their own errors already are, and that correlation is measurable from data supervisors collect.

5.4 The correlation budget

The remedy is legible only at population level. With K independently-erring models the correlated term becomes $w^2 \sigma_y^2/K$, and requiring the consensus to be no worse than the no-AI benchmark gives a *correlation budget*

$$K_{\min} = \frac{M \sigma_y^2}{\sigma_x^2 + 2\sigma_y^2} \approx M/4 \quad (6)$$

in our calibration, and never more than $M/2$ however poor the advisor. A hundred analysts leaning rationally on shared AI need twenty-five independent models before the advice stops degrading their collective judgment. Real models are not independent, so the binding constraint is *effective* diversity: at pairwise model-error correlation ρ_m we have $K_{\text{eff}} \leq 1/\rho_m$, so $K_{\min} = 25$ demands that model errors correlate at under 4%, which no population of frontier models trained on overlapping corpora plausibly satisfies. A better advisor needs a smaller budget ($M/13$ at $\sigma_y = 0.3\sigma_x$) but still leaves the consensus 7.6 $\times$ worse. Model quality does not solve a correlation problem.

This belongs to the family of externalities identified for precise public signals [23], but on weaker assumptions: there, agents overweight public information because they have a coordination motive; here they weight it exactly as Bayes requires, and the wedge comes purely from the shared draw. The experimental counterpart [11] and the monoculture result [20] report the same direction of effect, and supervisors treat it as a stability question [10].

6 From Teaming to Delegation under Agency

Our answer to what should replace teaming is not a new metaphor but an old formalism, and the interesting part is what breaks. In agency theory the agent has a reservation utility, can be incentivised, and bears residual risk. A model has none of these, so the incentive-compatibility half of the contracting apparatus has no counterpart here. The AI is not the agent; it is production and monitoring technology, and the expert remains the agent to a client-principal. That is why liability does not move, and why the informativeness principle applies to the model's outputs as signals rather than as acts. Three primitives survive, plus one the theory never had to consider.

Mandate. The contractible scope of the delegation. Aghion and Tirole's separation of formal from real authority [2] is what the field most needs: who holds *real* authority is an empirical question about who holds decision-relevant information, not a design declaration, and an expert with formal authority over a model she cannot audit has already ceded it. Fügner et al. find that human-AI pairs beat the standalone AI only when the AI delegates to the human [12], which is exactly the reallocation financial law forbids. The efficient and the lawful allocations point in opposite directions.

Monitoring technology. Holmström's informativeness principle [17]: condition on a signal iff it is incrementally informative about the agent's action. The calibration horizon does not say realised outcomes are uninformative, since weakly they are, but that their

signal-to-noise is so low that the optimal contract loads nearly all weight onto cheaper signals. That is the design implication, and the most actionable claim in this paper: stop calibrating on outcomes and *manufacture* informative ones. Concretely, this means prospective held-out evaluation, randomised withholding of advice so that a counterfactual exists at all, and process audits of the kind banking supervision calls “effective challenge”. Ben-Michael et al. [4] give the evaluation framework for the resulting three-way comparison; extending it to long-horizon continuous outcomes is open work.

Residual liability. Because the machine cannot bear risk, liability does not merely happen to stay with the human: it *cannot* move, and is not bargainable. The SEC’s interpretation makes the adviser’s fiduciary duty non-waivable, and the EU AI Act requires natural-person oversight of high-risk systems [8]. Supervisory guidance has meanwhile become less prescriptive: the 2011 model-risk guidance naming a model owner with ultimate accountability was superseded in April 2026 by guidance a third its length, which drops the term and declines to scope generative and agentic AI at all. Elish’s moral crumple zone describes blame concentrating on the operator nearest the system [7]; regulated finance inverts it, since a named fiduciary exists and shared-agency language diffuses responsibility away from her, which is the documented limitation of oversight that legitimates without constraining [14]. To the question of what work the construct does, and for whom: it serves the institution, for which responsibility becomes more diffuse, more than it serves the expert, who retains the liability without the authorship.

Correlation budget. $K_{\min}$. Classical agency theory is bilateral and has no place for it, no dyadic vocabulary can express it, and it is the primitive most obviously missing from practice.

6.1 What the prescription costs

A design recommendation should carry a price, so here is ours. Eq. 1 assumes the ideal paired design in which the difference portfolio is directly observed. A real withholding programme instead compares two disjoint sub-books, with advice on for a fraction p of decisions and off for the rest. For a two-sample split the standard error carries a factor $1/\sqrt{Np(1-p)}$, so the requirement becomes $N^*/p(1-p)$ expert-years, which is 785 at $\Delta IR = 0.20$ and is minimised at $p = \frac{1}{2}$: 15.7 years for a 50-manager firm, 3.9 for a 200-manager firm, under one for a thousand-manager firm. Withholding from only a fifth of decisions costs a further $1.56\times$.

Two things follow. The programme is costly but feasible: four years is a governance cycle rather than a career, and it is the difference between a supervisable claim and an unsupervisable one. But it is available only at scale, so the institutions that can identify whether their AI advisors help are also those that benefit most from the diffusion of responsibility that teaming language supplies. Capacity to measure the effect and the weakest incentive to look for it reside together.

7 Limitations

The models are stylised, deliberately. Both are identification arguments rather than empirical estimates: one establishes that a quantity is not recoverable from the evidence an expert receives, the other that a population externality exists under Bayesian individual

behaviour. Neither estimates the size of the effect in any particular firm, and the parameters in Table 1 are order-of-magnitude judgements: a reader who disputes a row should recompute rather than accept it.

Scope and generality. Our argument concerns judgment under low validity and assigned liability. It does not apply to the large share of professional work that is content creation with fast, gradeable feedback, where measured gains are real and teaming may describe the work well [26]; nor do we claim collaboration is unhelpful in general. Finance is also an extreme case, and the best-instrumented one. We expect P2 and P3 to bind wherever outcomes are noisy and liability is personal, including hiring, parole and long-horizon clinical decisions, but we have not shown it: the non-finance rows of Table 1 are illustrative, and computing τ in a second instrumented domain is the obvious next study.

The strongest counter-position. A complementarity framework can absorb our result as a low-complementarity domain it already characterises [13]. We accept the diagnosis and reject the sufficiency of the prescription: a framework that tells you complementarity is unavailable leaves the design question open in exactly those domains. And once evaluation moves to held-out benchmarks, randomised withholding and process audits, the object being designed is a delegation contract with a monitoring technology.

What would change our mind. Three observations would. A demonstration that assistance adds $\Delta IR \geq 0.44$ to a discretionary book would put calibration back inside a career. Evidence that experts form reliance from audited process signals rather than surface cues would make the non-satisfaction of P2 immaterial. And unaided expert forecasts correlating above $\rho = 0.25$ would reverse the sign of Study 2 in that population. All three are measurable with data large institutions already hold.

8 Conclusion

The teaming vocabulary works where three conditions hold: the team can see what its own decisions produced, it can see this quickly, and responsibility is genuinely shared. Expert-AI financial judgment meets none of them, and we have tried to establish this by calculation rather than assertion. A portfolio manager would need two centuries of her own record to establish that an AI advisor helps; her employer could do it in four by withholding advice at random; experts sharing a model are each right to rely on it while their collective judgment worsens; and liability stays with a named person, because a model has no assets and cannot bear risk. What remains once teaming is set aside is delegation under agency, with four contractible primitives: mandate, monitoring technology, residual liability, and correlation budget. Finance has used the first three since the 1970s; shared models now require the fourth. Where τ differs by orders of magnitude between the unit that forms trust and the one holding the data, advice to calibrate is addressed to someone who cannot act on it.

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